

01_FOG_SPEED_GOVERNOR_ARCHITECTURE

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EVIDENCE PROVENANCE: [MATHEMATICAL DERIVATION]

Closed-form physical derivation of stopping distance, retarder thermal limits, and tire-road friction envelopes.

1. The 5-Constraint Governor Equation

Governed safe speed is computed as $v_{safe} = \min(v_{stop}, v_{retarder}, v_{traction}, v_{curve}, v_{mine})$. Each term represents an inviolable physical boundary.

- **1. Stopping Sight Distance (v_{stop}):** Ensures vehicle can halt within current optical visibility V_{vis} : $S_{stop} = v * \tau_{total} + \frac{v^2}{2 * a_{dec}} \leq V_{vis}$.
- **2. Retarder Thermal Limit ($v_{retarder}$):** Gravitational power on downhill ramp must not exceed 1,119 kW retarder capacity.
- **3. Traction / Slip Limit ($v_{traction}$):** Deceleration bounded by tire-road friction μ on wet clay/slurry.
- **4. Curve Stability (v_{curve}):** Centripetal speed on 22m radius hairpin limited by $v = \sqrt{\mu * g * R}$.
- **5. Statutory Limit (v_{mine}):** DGMS regulated maximum speed (40 km/h nominal / 0.80 m/s bench baseline).

2. Parameter Nomenclature

Parameter	BEML BH100 Full Scale	Bench Prototype Scale	Classification
Vehicle Mass (M)	165,500 kg (Loaded)	1.85 kg chassis	MEASURED
Reaction Time (τ)	1.85 s (Driver+Hydraulic)	0.082 s (Debounce+Wi-Fi)	MEASURED
Road Friction (μ)	0.28 to 0.35 (Wet Slurry)	0.35 (Rubber on Bench)	MEASURED
Ramp Gradient (G)	8.0% to 12.0% downhill	8.0% to 12.0% tilt bench	GEODATA
Hairpin Radius (R)	22.0 m centerline	0.80 m bench arc	GEODATA